

198888 PR Data Analysis Lab:

Applied Project Report

Identifying the drivers of employee satisfaction in text: A comparative analysis of an LLM- and dictionary-based approach on Kununu and Glassdoor reviews

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1. **Introduction**

Employee satisfaction is one of the most researched topics in organizational psychology (Nerdinger, 2019). Most instruments that measure employee satisfaction and related constructs rely on self-reports of participants. Unstructured data from employee-review platforms, however, offers a new data source with certain advantages over survey-based methods (Septiandri et al., 2025). Natural language processing (NLP) has been used to extract meaning from such data and provides a promising way to analyze large amounts of anonymous online text. With the rise of large language models (LLMs), new possibilities for NLP have emerged: when classifying text, LLMs reach a high accuracy that can even exceed that of professional human annotators (Gilardi et al., 2023). This project compares two methods for classifying employee-satisfaction data. Rather than searching inductively for patterns, it builds on an established psychological theory called the Motivator-Hygiene-Theory by Frederick Herzberg (1959) to test whether this theory still holds as a framework for employee satisfaction in the modern workspace. The Motivator-Hygiene-Theory states that satisfaction and dissatisfaction should not be seen as opposite ends of a single spectrum but are instead driven by two distinct sets of factors (Herzberg, 1959).

Table 1
Motivation and Hygiene Factors by F. Herzberg (1959)

Motivation Factors	**Hygiene Factors**
Achievement	Company Policy & Administration
Recognition	Supervision
Advancement & Promotion	Working Conditions
Work Itself	Personal life
Growth	Interpersonal Relations
Responsibility	Job Security
	Status

Note. Research suggests that Salary & Benefits acts as a Motivator depending on context and can therefore not be categorized (Ozsoy, 2019).

This implies that the reviews of satisfied employees address different themes than those of dissatisfied employees. All factors are listed in Table 1. The two methods are tested for their ability to classify reviews into this predefined framework. The first is a dictionary-based approach that matches reviews against factor-specific word lists. The second method uses an LLM to classify each individual review. The dictionary-based approach can be seen as the simplest form of NLP: it lacks any understanding of context but is fully reproducible and

traceable. Using an LLM, in contrast, introduces the black-box problem. It is nonetheless a very promising technique, since it can capture context such as irony.

2. Data & Methods

This chapter describes the two datasets that were used in this project as well as the preprocessing and statistical methods that were applied to classify and compare the reviews.

2.1 Data Source

The platform kununu.com was scraped. 38 of the 40 German DAX-companies were chosen. 2 of the 40 companies could not be found on kununu. Additionally, 8 non-DAX companies were chosen to widen the spectrum of industries. This resulted in a dataset containing 19.370 observations, at a maximum of the most recent 500 reviews per company. The included companies and the count of reviews per company are displayed in Figure 1. For later comparison another dataset was downloaded from the platform kaggle.com. The data was scraped from Glassdoor and published on Kaggle by the author David Gauthier (Gauthier, 2024). 114.325 out of 836.254 reviews were used after cleaning and due to price limitations of the LLM-based classification. For this project two fields are relevant: “pros” (Glassdoor) / ”what do you like about your employer” (kununu) and “cons” (Glassdoor) / ”what do you dislike about your employer” (kununu). The two data sources differ in multiple ways. Firstly, the timeline deviates: the Glassdoor dataset only contains reviews that were published up until 2021, meanwhile the kununu dataset includes the most recent 500 reviews of each company starting from 06.06.2026 (Figure 2). The datasets also differ in region and language: while only English-language Glassdoor reviews were used, the kununu dataset contains German-language reviews.

Figure 1
Number of reviews per company in the kununu dataset.

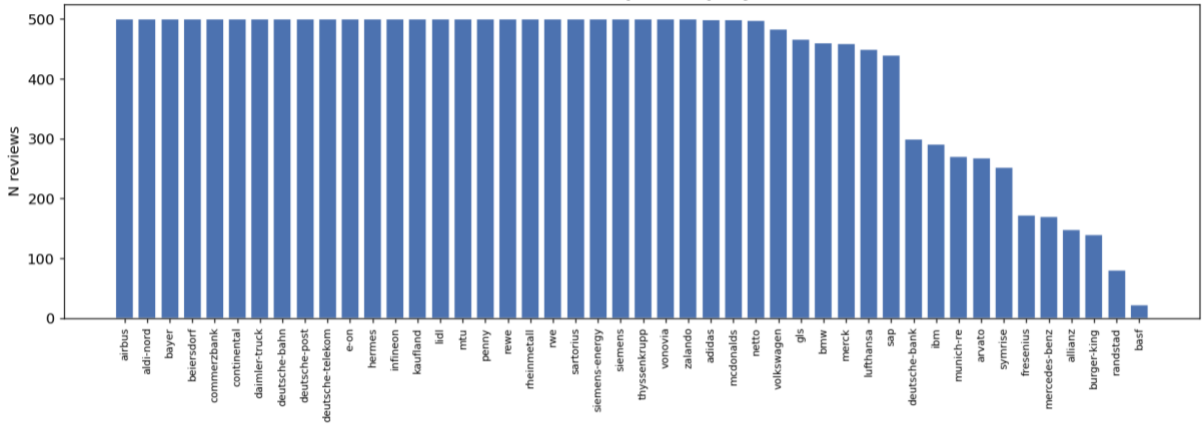
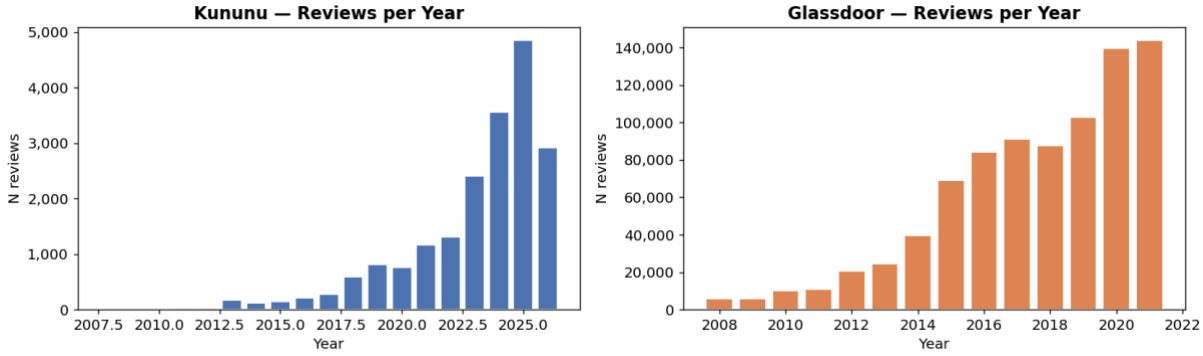


Table 2
Descriptive Information about the two samples.

	Kununu	Glassdoor
N reviews	19.370	836.254 (114.325 classified)
N companies	46	428
Year range	2008–2026	2008–2021
Mean rating (1–5)	3.57 (SD 1.15)	3.66 (SD 1.17)
Pros filled	12.710 (66%)	836.252 (100%)
Cons filled	11.056 (57%)	836.241 (100%)
Median words (non-empty)	Pros 6 | Cons 8	Pros 9 | Cons 9
Language	German	English

Figure 2
Reviews per year by platform.



2.2 Preprocessing and Postprocessing

The Glassdoor data contained reviews written in multiple languages. Due to this, language detection was applied using the *langdetect* library and only reviews identified as English (lang = "en") were retained. This reduced the raw input from 838.566 to 836.254 English-speaking reviews.

A post-processing step was applied to the kununu LLM classifications to correct for a systematic model artefact. When presented with a meaningless text, which only indicated that everything was fine (such as *"alles"*), the model wrongly assigned all 14 factors a value of 1. To correct for this, any review where all 14 factor labels were set to 1 and the corresponding text was shorter than 15 characters was reset to all zeros, coding that no factor was mentioned. This criterion affected 1.783 pros entries and 3.782 cons entries.

2.3 Statistical Methods

Cross-platform Comparison. To compare Glassdoor and kununu reviews, a Spearman rank-test was conducted to test if the importance of factors is aligning between the two platforms. This was especially relevant since kununu had 13 additional open-text fields that Glassdoor

was missing. In Glassdoor all open-text answers are canalized into the pros- and cons-fields. Since the project aims to measure and compare the expression of the 14 factors inside the pros and cons fields, it is crucial that employees use these fields to address the factors even if there is an alternative way to speak up about them.

Classification Quality. Next to the two classification methods that were already mentioned, there was a third method used as a gold standard to measure classification accuracy. For this, the author classified a subsample of 238 reviews into the same factor-schema as the other two approaches. To compare the goodness of classification, Cohen's κ was calculated.

Prevalence Bias. The κ values in the analysis above capture whether the methods agree with human coding, but they do not reveal the direction of disagreement. To address this, I computed the prevalence bias for each factor, which is defined as the percentage of reviews coded as positive by the method minus the percentage coded as positive by the human coder. A positive bias indicates overclassification, a negative bias underclassification.

3. Results

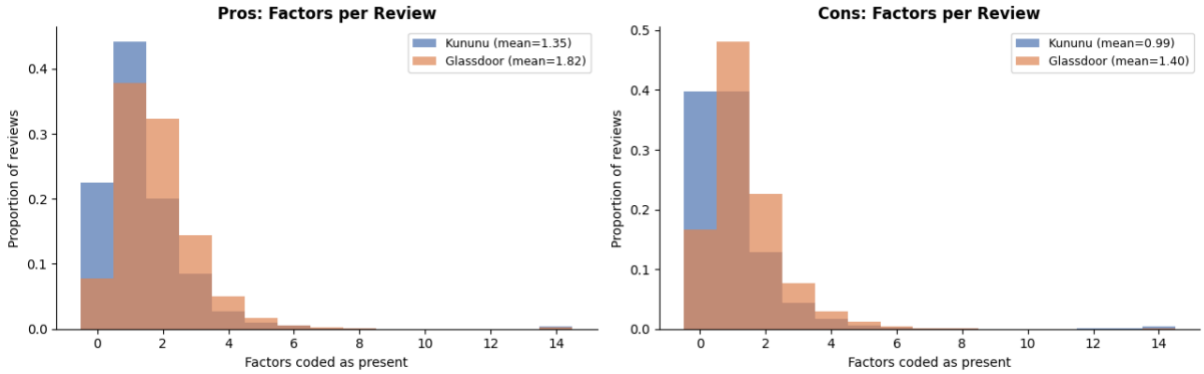
This chapter presents the results of the analysis.

3.1 Cross-Platform Comparison

Across the platforms, the ranking of factor mention frequency is similar, with a Spearman rank correlation of ρ = .804. Overall, Herzberg Factors were mentioned more frequently in the Glassdoor sample (Figure 3). This could be because reviewers in the Glassdoor sample used more words (median word count 9 vs. 7; Table 2). It could also stem from differences in UX/UI design, where kununu provides separate categories that overlap with some of the factors.

Figure 3

Number of Herzberg Factors Mentions per Review by platform



Note. In Pros, Glassdoor reviews mention +0.47 times more Herzberg Factors than kununu reviews. In Cons, Glassdoor reviews mention +0.41 times more Herzberg Factors than kununu reviews.

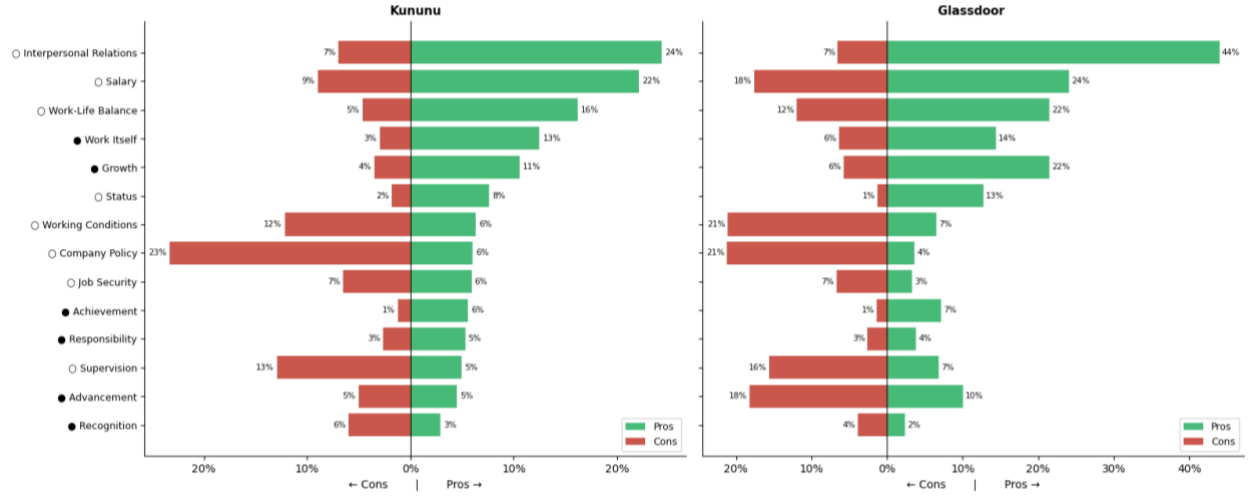
Especially for the factors Interpersonal Relations (pros) and Salary (cons), Growth (pros), Working Conditions (cons) and Advancement (both), Glassdoor mentions substantially outnumber kununu mentions, and all these factors correspond to separate kununu categories. To connect the factors back to the theory, it is of interest whether employees talk about them more positively or negatively. The direction (whether a factor is mentioned more positively or negatively) is also consistent across both platforms. Advancement is the only exception: on Glassdoor it was mentioned more often positively, whereas on kununu it appeared equally in pros and cons.

Table 3
Spearman rank correlation between Kununu and Glassdoor samples.

All 28 (14 factors × Pros/Cons)	ρ = 0.804
Pros only (N = 14)	ρ = 0.798
Cons only (N = 14):	ρ = 0.833

Note. ρ = Spearman-Rank Coefficient.

Figure 4
Relative Frequency of mentions per Herzberg-factor in Pros and Cons by platform.



Note. Black dot = Hygiene Factor. White dot = Motivator.

3.2 Classification Quality

To evaluate how well the two automated methods capture Herzberg factors, I compared their classifications against a human-coded gold standard. A random sample of 238 kununu reviews was manually coded, and Cohen's κ was used to measure agreement between each automated method and the human coder. κ accounts for agreement occurring by chance and ranges from 0 (no agreement beyond chance) to 1 (perfect agreement).

Across all 14 factors, the LLM achieved a mean κ of 0.54 and the dictionary method a mean κ of 0.40 (Landis & Koch, 1977: 0.40–0.60 = moderate, 0.60–0.80 = good). When restricting

the comparison to the factors that appeared in more than 10 instances in total (in pros and cons together) in the human coding, the LLM performance reached $\kappa = 0.59$ and the dictionary remained at $\kappa = 0.40$ (Table 4). These two factors (Achievement and Recognition) were excluded since they only appeared 6 and 9 times, respectively, in the human-classified sample. With so few instances, their κ values would rest on too little data to be reliably interpretable.

Table 4
Pairwise Agreement Between the Classification Methods and Human Coding

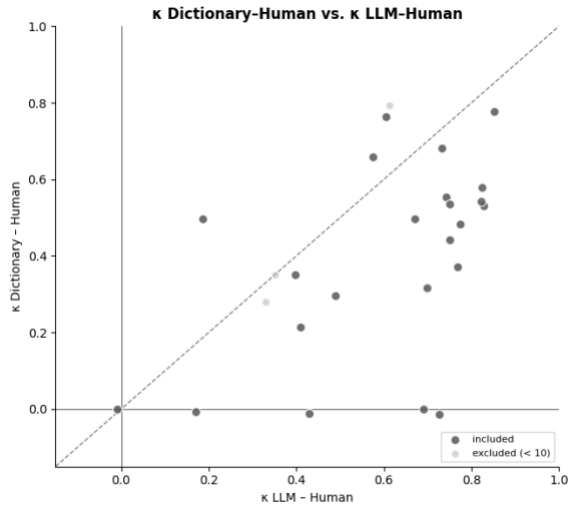
	Dictionary	**LLM**	**Human**
Dictionary	1.00		
LLM	0.33	1.00	
Human	0.40	0.54	1.00

Note. Each cell shows Cohen's κ between two approaches.

Factor-level results show a pattern: both methods perform best on concrete, tangible factors such as Salary (κ_LLM = 0.79) and Interpersonal Relations (κ_LLM = 0.75), where the presence of a factor is signaled by specific, unambiguous words. Agreement is lower for more abstract or inferential factors such as Status (κ_LLM = 0.34). These results suggest that the LLM approach is a viable method for large-scale classification of employee reviews, particularly for common and concrete Herzberg factors. The dictionary approach, while fully transparent and deterministic, systematically underperforms, especially on factors that cannot be reliably detected through keyword lists alone (Figures 5 and 6).

Figure 5

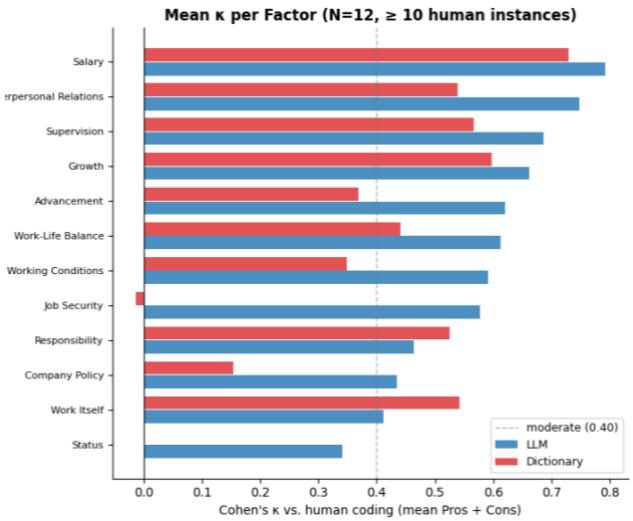
Inter-rater Agreement with Human Coding: Dictionary vs. LLM



Note. Each point is a factor x pro/con combination. Points on the diagonal indicate equal agreement for both methods; points above the diagonal indicate higher Dictionary agreement. Grey points (Achievement, Recognition) were excluded from the summary due to fewer than 10 human-coded instances.

Figure 6

Cohen's κ Against Human Coding by Factor



Note. Bars show mean κ averaged over Pros and Cons. Factors with fewer than 10 human-coded instances (Achievement, Recognition) are excluded.

3.3 Prevalence Bias

Table 5

Prevalence Bias of the LLM and Dictionary Methods by Herzberg Factor

Factor	% Human	Bias LLM	Bias Dict
Salary	22.9	-2.3	-3.6
Advancement	8.0	-1.1	-5.7
Company Policy	14.7	-0.2	-11.8
Interpersonal Relations	18.5	+0.6	-6.3
Responsibility	3.6	+1.1	-1.9
Work-Life Balance	13.4	+1.3	-7.1
Supervision	7.4	+1.7	-3.8
Status	2.7	+1.9	-2.7
Working Conditions	8.4	+2.1	+1.7
Work Itself	6.7	+2.9	-4.0
Growth	5.9	+3.2	-2.5
Job Security	2.9	+3.6	-2.1
Mean		+1.2	-4.2

Note. Bias in percentage points (% method − % human). Positive values indicate overclassification, negative values underclassification. The percentages next to the factor labels show the prevalence in the human-coded sample.

To reveal the direction of disagreement, the prevalence bias for each factor was calculated (Table 5). The LLM shows a slight positive bias on average (+1.2 percentage points), with no consistent directional pattern across factors. The bias is largest for rare categories: Job Security (+3.6 pp), Growth (+3.2 pp), and Work Itself (+2.9 pp) are systematically overestimated. However, these estimates should be interpreted with caution: factors with few

human-coded instances are precisely those where the human baseline is most sensitive to sampling variability. Work Itself, for example, was not coded a single time in the cons of the human subsample, meaning any LLM positive classification for that cell mechanically produces a positive bias regardless of classification quality. The overestimation in these categories may therefore reflect the limits of the 238-review validation sample rather than a systematic tendency of the model.

The dictionary method displays a qualitatively different pattern: a consistent negative bias across nearly all 12 included factors (mean = −4.2 pp), indicating systematic underdetection. Working Conditions (+1.7 pp) is the only factor where the dictionary slightly overestimates.

4. Discussion

The results suggest that the LLM-based approach is a practicable method to classify large amounts of employee reviews. With a mean κ of 0.54 across all factors, and 0.59 when the very rare factors are excluded, it reached moderate agreement with the human coding. As mentioned before, the agreement was highest for concrete factors such as Salary ($\kappa = 0.79$) and Interpersonal Relations ($\kappa = 0.75$) and lowest for more abstract factors such as Status ($\kappa = 0.34$). This suggests that the method works best when a factor is signaled through specific words. For the goal of analyzing large amounts of employee review data, this level of agreement seems sufficient, especially since the dictionary approach, which is the more transparent alternative, performed worse (mean $\kappa = 0.40$). Besides that, the dictionary method showed a consistent negative bias (mean = −4.2 pp) and therefore underdetected nearly every factor, whereas the LLM showed only a slight positive bias (+1.2 pp) without a clear directional pattern. Since its positive bias was mostly found in factors with very few human-coded instances, where one single inferred mention already produces a large deviation, this pattern can rather be seen as an effect of the small validation sample than as a real tendency of the model to over-classify.

Besides the comparison of the methods, the high rank correlation of the factor structure between the two platforms ($\rho = .804$) is also of interest. The kununu and Glassdoor samples differ in language (German vs. English), region, time frame and the design of the review-process and structure, but the importance of the Herzberg Factors was still very consistent between them.

There are some limitations that must be kept in mind. First, the gold standard was coded by only one person. Without a second coder the reliability of the human baseline itself cannot be estimated. As mentioned before, the small sample also makes the bias estimates for the rare

factors unstable. Next, the dictionary was used as a fixed wordlist, so an adapted or larger dictionary could possibly reduce the gap to the LLM. Additionally, other classification methods such as word-embeddings should be compared to the LLM-based approach. Finally, for this analysis a rather fast and superficial model, Gemini 2.5 Flash, was used. For future research it would be of interest to use a more powerful model, since it seems likely that this would further increase the classification accuracy.

5. Conclusion

Overall, this project shows that an LLM can classify employee reviews into the Herzberg Factors with an acceptable accuracy, while the dictionary approach lags behind, especially for the broader factors. The consistent factor structure across kununu and Glassdoor also suggests that the Motivator-Hygiene-Theory still works as a useful framework for employee satisfaction in the modern workspace.

Literature

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